



INSTITUTE OF MEDICINE

SM00146, Systems Biology and Omics Technology, 1.5 credits

Systembiologi och omiska teknologier, 1,5 högskolepoäng

Third-cycle level / Forskarnivå

Confirmation

This syllabus was confirmed by the Council for PhD Education at Sahlgrenska Academy on 2026-03-05, and is valid from Autumn semester 2026.

Responsible Department

Institute of Medicine, Sahlgrenska Academy

Entry requirements

Registered as a doctoral student.

Learning outcomes

After completion of the course the PhD student is expected to be able to:

Knowledge and understanding

Explain the concept and properties of different next-generation sequencing (omics) data

Competence and skills

Construct a basic workflow for omics analysis, including data exploration and visualization.

Judgement and approach

- Critically evaluate and discuss recent advancements in systems biology, focusing on their state-of-the-art applications in biomedical research

- Identify and justify the choice of omics data types and analytical approaches for use in their own research.

Course content

The course introduces the fundamental principles and practical applications of systems biology and omics analysis, focusing on transcriptomics, metabolomics, proteomics, and genomics. Through hands-on lab sessions, group presentations, and discussions of recent research, participants will gain both conceptual and practical understanding of omics-driven approaches in modern biomedical science. The course emphasizes critical evaluation of next-generation sequencing data, construction of basic analysis workflows in Python, and interpretation of data in a biological context. Designed for experimental and molecular biologists, it aims to bridge the gap between experimental and computational research, helping participants integrate omics strategies into their own projects and collaborate effectively across disciplines.

Types of instruction

Lectures, workshops, and discussions

Language of instruction

The course is given in English

Grades

The grade Pass (G) or Fail (U) is given in this course.

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Types of assessment

1. Oral Group Assignment
2. Written individual assignment

If a PhD student, who has been failed on the same examining course component twice, requests a change of examiner before the next examination session, a request of this kind should be sent in writing to the department responsible for the course, and granted, unless there are special reasons to the contrary (Chapter 6, Section 22, Higher Education Ordinance).

Course evaluation

The course evaluation will be provided both as a written questionnaire (joint for Sahlgrenska Academy) and orally as a discussion between the students and the course leader(s). The course

responsible teacher compiles analysis of the course evaluation and makes suggestions for further development of the course. The result and any changes in the set-up of the course shall be communicated both to the students that carried out the evaluation and to the students who are about to start the course.

Other information

Access to a computer (laptop) and the internet is required. No specific requirements for the computers.